WEEK 9 · LECTURE L1 · UNIT III

Algorithmic Decision-Making in HR

Where algorithms now make or shape HR decisions, and what the Amazon case teaches.

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AFTER THIS LECTURE YOU SHOULD BE ABLE TO

What this lecture prepares you for

- Identify the HR decisions algorithms now shape, and the specific risk in each.
- Explain how a model trained on past decisions inherits their bias.
- Explain why equal precision can coexist with very unequal error rates.
- Apply Kellogg et al., Tambe et al., algorithm aversion and algorithm appreciation.

COURSE OUTCOMES

CLO 3 (algorithmic decision-making in recruitment, appraisal and promotion).

WHERE ALGORITHMS ACT

Five HR decisions algorithms now shape

Decision	What the algorithm does	The risk
Screening	Ranks or filters applicants	Learns past bias; unknown validity
Matching	Suggests roles, shifts or tasks	Feedback loops that lock people in
Appraisal	Scores performance from activity data	Measures what is easy, not what matters
Promotion	Ranks readiness	Proxies for tenure, gender or location
Attrition prediction	Flags who may leave	False alarms that change how people are treated

DASTIN (2018): THE AMAZON CASE

A model that learned the past

What happened

Amazon built an experimental tool to score resumes and scrapped it after finding it downgraded resumes mentioning women's activities.

Why

It was trained on about ten years of resumes from mostly male hires, so it treated male-linked signals as merit.

Why fixes failed

Removing gender words did not help: other features acted as proxies.

A model trained on past decisions inherits their bias, even if no one intended it.

AN ILLUSTRATIVE FLIGHT-RISK MODEL

Equal precision, unequal errors

Measure	Group A (200 staff)	Group B (200 staff)
Actual leavers (base rate)	40 (20%)	30 (15%)
Flagged high risk	100 (50%)	40 (20%)
Precision (flags that are right)	25%	25%
False positive rate (stayers wrongly flagged)	47%	18%

EXAM LENS

Illustrative figures. The model is equally precise for both groups, yet flags Group A far more often and wrongly flags nearly half its stayers. Always examine error rates by group, not overall accuracy.

THEORY IN DEPTH · KELLOGG, VALENTINE AND CHRISTIN (2020)

The six Rs of algorithmic control

THE CORE IDEA

Algorithms control work through three mechanisms: direction, evaluation and discipline.

HOW IT WORKS

Restricting and recommending (direction); recording and rating (evaluation); replacing and rewarding (discipline).

LIMITS AND CRITIQUE

Workers resist through what the authors call "algoactivism", so control is never complete.

IN PRACTICE

Gig platform allocation scores restrict, rate and reward workers without a visible manager.

EXAM LENS

Classify each feature of an algorithmic HR system into the six Rs; then discuss worker resistance.

THEORY IN DEPTH · TAMBE, CAPPELLI AND YAKUBOVICH (2019)

Four reasons AI is hard in HR

THE CORE IDEA

HR outcomes are complex to measure, data sets are small, fairness and law constrain models, and employees react to being judged.

HOW IT WORKS

They recommend causal reasoning, randomized experiments where possible, and involving employees in design.

LIMITS AND CRITIQUE

Written before large language models changed what AI can do with text.

IN PRACTICE

Most organizations' HR data sets are small; a model trained on them will be unstable for small groups.

EXAM LENS

The four challenges are an efficient structure for any question on AI in HR.

THEORY IN DEPTH · DIETVORST, SIMMONS AND MASSEY (2015)

People abandon algorithms after one mistake

THE CORE IDEA

Algorithm aversion: after seeing a model err, people prefer human judgment even when the model is more accurate.

HOW IT WORKS

People forgive human errors more readily than machine errors.

LIMITS AND CRITIQUE

Letting people slightly adjust a model's output reduces aversion.

IN PRACTICE

Managers who see one bad flight-risk flag may stop trusting all analytics, good and bad alike.

EXAM LENS

Pair with Logg et al. to show trust in algorithms can err in both directions.

THEORY IN DEPTH · LOGG, MINSON AND MOORE (2019)

People also over-trust algorithms

THE CORE IDEA

Algorithm appreciation: lay people often weight advice more when told it comes from an algorithm.

HOW IT WORKS

Appreciation is stronger for numeric estimates and weaker among experts.

LIMITS AND CRITIQUE

Aversion and appreciation coexist; which wins depends on the task and the person.

IN PRACTICE

A flight-risk score shown as a precise number may be trusted more than it deserves; show its error rate beside it.

EXAM LENS

A strong answer reconciles aversion and appreciation rather than citing only one.

PRACTICE FOR THE END-TERM EXAMINATION

Questions to rehearse

"Algorithms in HR do not remove bias; they encode and scale it." Critically evaluate with reference to the Amazon case and Tambe et al. (2019). **20 marks**

A strong answer: explains learning from biased labels and proxies; uses Tambe et al.'s four challenges; acknowledges potential consistency benefits; proposes safeguards.

Explain algorithm aversion and algorithm appreciation, and discuss their implications for how HR presents algorithmic recommendations to managers. **20 marks**

A strong answer: explains both findings and moderators; proposes design: show error rates, allow limited adjustment, retain accountability.

Explain Kellogg et al.'s six Rs of algorithmic control. **10 marks**

A strong answer: names the six Rs under direction, evaluation and discipline with examples.

CARRY THESE INTO THE EXAMINATION

Three things to remember

1. **Every HR algorithm carries a specific risk; name it**
2. **Models trained on past decisions inherit past bias**
3. **Precision can be equal while error rates differ sharply**

SET READINGS · FROM THE LESSON PLAN

Read for this lecture

- Tambe, P., Cappelli, P., & Yakubovich, V. (2019). *California Management Review, 61*(4), 15–42.
- Kellogg, K. C., Valentine, M. A., & Christin, A. (2020). *Academy of Management Annals, 14*(1), 366–410.

COMPANION

Each set reading is explained in the matching Unit Reader on rksingh.tech/materials.

THEORY SOURCES · FOR REFERENCE

Sources for this lecture

Works named on the slides of this lecture. They are background sources, not set readings.

Dastin, J. (2018, October 10). Amazon scraps secret AI recruiting tool that showed bias against women. *Reuters*.

Dietvorst, B. J., Simmons, J. P., & Massey, C. (2015). Algorithm aversion: People erroneously avoid algorithms after seeing them err. *Journal of Experimental Psychology: General, 144*(1), 114–126. https://doi.org/10.1037/xge0000033

Kellogg, K. C., Valentine, M. A., & Christin, A. (2020). Algorithms at work: The new contested terrain of control. *Academy of Management Annals, 14*(1), 366–410. https://doi.org/10.5465/annals.2018.0174

Logg, J. M., Minson, J. A., & Moore, D. A. (2019). Algorithm appreciation: People prefer algorithmic to human judgment. *Organizational Behavior and Human Decision Processes, 151*, 90–103. https://doi.org/10.1016/j.obhdp.2018.12.005

Tambe, P., Cappelli, P., & Yakubovich, V. (2019). Artificial intelligence in human resources management: Challenges and a path forward. *California Management Review, 61*(4), 15–42. https://doi.org/10.1177/0008125619867910